Conditional Meta-Learning for Global-to-Local Prediction of Population Genetic Diversity and Climate Vulnerability

Abstract

Genetic diversity underpins populations' resiliency to environmental change, yet it remains poorly mapped because genetic sampling is often sparse and spatially uneven. We developed a Conditional Neural Process (CNP) meta-learning framework to downscale sparse genetic observations into fine-resolution, spatially continuous predictions of population genetic diversity with quantified uncertainty. The model was pre-trained on 19,163 populations of 1,108 species globally and fine-tuned on 461 populations from 16 California coastal species, jointly modeling expected heterozygosity (He) and nucleotide diversity (π). Fine-scale predictors integrated 1-km terrestrial climate, marine environmental conditions, and land use pressure. Within-species leave-one-out evaluation showed strong predictive performance for He and π, while resolving within-species spatial variation (median per species) and outperforming inverse-distance weighting. Under SSP5–8.5 projections for 2081–2100, changes in genetic diversity were modest, mixed in direction, and species-specific, with little change in community-mean diversity. Vulnerability reflected the joint spatial patterns of standing genetic diversity and projected climate exposure rather than exposure alone. Our framework provides a practical approach for downscaling sparse genetic data into fine-resolution diversity maps, supporting the integration of population genetic variation into spatial conservation and climate vulnerability assessments.

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1. Introduction

Genetic diversity is a fundamental component of biodiversity, underpinning populations' capacity to respond to environmental change, maintain evolutionary potential, and persist under increasing anthropogenic pressures (Frankham, 2005; Hughes et al., 2008; Hoban et al., 2020). As climate change, habitat fragmentation, biological invasions, and land-use change reshape ecosystems worldwide, conserving genetic diversity has become an increasingly important objective of biodiversity management and climate-adaptation strategies (Sgrò et al., 2011; Razgour et al., 2019; Hoban et al., 2021). Populations with greater genetic diversity can have greater capacity to respond to environmental change and reduced extinction risk (Reed & Frankham, 2003; Frankham, 2005; Forsman, 2014), making genetic information increasingly relevant to conservation planning, habitat restoration, connectivity assessments, and identification of climate refugia (Carroll et al., 2014; Hoffmann et al., 2015; Hoban et al., 2020). Spatially explicit information on population genetic diversity is therefore needed to identify where evolutionary potential may be concentrated or depleted and how these patterns intersect with future environmental change.

Yet genetic diversity remains poorly mapped at the spatial scales required for conservation planning. Advances in molecular technologies and large-scale sequencing have

greatly expanded the availability of population genetic data (Allendorf et al., 2010; Hohenlohe et al., 2021), but sampling remains strongly uneven among species, regions, and taxonomic groups (Leigh et al., 2021; Theodoridis et al., 2021). Most population genetic studies focus on individual species and a limited number of sampled populations, leaving genetic diversity represented by geographically sparse point observations rather than spatially continuous information (Manel et al., 2003; Schmidt et al., 2020). Environmental association models, spatial interpolation, and machine-learning approaches can extend inference beyond sampled locations (Muff et al., 2019; Razgour et al., 2019; Lawrence et al., 2019), but are commonly developed independently for individual species. Consequently, information accumulated across thousands of populations and species globally remains difficult to transfer to data-limited species within a particular region.

Genetic sampling limitations create a spatial mismatch in conservation genetics. Conservation decisions are often implemented across landscapes and multiple species, whereas genetic diversity is typically measured from a limited number of populations and analyzed one species at a time (Manel et al., 2003; Hoban et al., 2021; Schmidt et al., 2020). Such sparse and taxonomically uneven sampling constrains the development of spatially continuous genetic-diversity information for multispecies conservation planning (Leigh et al., 2021; Theodoridis et al., 2021). Moreover, local genetic datasets are often too limited to train flexible predictive models independently for each species, particularly when the objective is to resolve fine-scale spatial heterogeneity. This data limitation is especially relevant to modern machine-learning approaches, which generally benefit from substantially larger training datasets (Allendorf et al., 2010; Leigh et al., 2021). An alternative is to leverage the growing body of population genetic information available across species and regions as a source of transferable knowledge; by learning shared diversity–environment relationships across many species and subsequently adapting them with limited observations from focal species

Although deep learning has transformed many fields by leveraging large training datasets to learn complex environmental relationships, the amount of genetic data available within a single management landscape is often insufficient to train robust deep learning models. This creates a fundamental challenge for conservation genetics: the spatial scale at which management decisions are made is often much larger than the scale at which genetic diversity is measured, while the amount of local training data remains too limited for modern deep learning approaches (Allendorf et al., 2010; Leigh et al., 2021). Consequently, methodological frameworks are needed to leverage information beyond local datasets and translate sparse genetic observations into continuous, landscape-scale predictions to inform conservation and climate adaptation planning.

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Advances in transfer learning and meta-learning provide a framework for accomplishing this task (Pan & Yang, 2010; Yosinski et al., 2014). Rather than fitting each species independently, meta-learning treats species as related prediction tasks and learns representations that can be shared across tasks while adapting to limited species-specific observations. A global-to-local framework could downscale sparse population genetic observations into fine-resolution, multispecies maps, while retaining species-specific responses to local environmental conditions.

Neural Processes (Garnelo et al., 2018) are particularly well suited to this setting. They learn distributions over functions conditioned on a variable number of context observations, allowing sparse observations from a focal species to inform predictions at unsampled locations. Related approaches, including Attentive Neural Processes (Kim et al., 2019), Matching Networks (Vinyals et al., 2016), and model-agnostic meta-learning (Finn et al., 2017), demonstrate how information learned across tasks can improve prediction when observations for

individual tasks are limited. Feature-wise Linear Modulation (FiLM; Perez et al., 2018) further allows shared representations to be conditioned on different genetic-diversity metrics. Together, these developments offer an opportunity to move from species-by-species modeling toward a unified framework for multispecies population genetic prediction.

Here, we developed a Conditional Neural Process meta-learning framework that learns population genetic diversity–environment relationships from a global database and transfers this information to generate fine-resolution predictions for multiple species within a regional conservation landscape. We use the California coast as a case study because it combines substantial environmental heterogeneity and conservation importance with expanding population-genetic resources. In addition, California's biodiversity and climate-adaptation initiatives create a direct need for spatially explicit information to support regional conservation planning (Gleason et al. 2023; Fielder et al. 2022). Rather than predicting a single species or genetic metric, our framework jointly models population genetic diversity across multiple species and accommodates both expected heterozygosity (H_e) and nucleotide diversity (π). Global pre-training provides transferable ecological representations, whereas regional fine-tuning integrates high-resolution terrestrial climate, marine environmental conditions, and land use pressure to downscale sparse genetic observations into spatially continuous predictions at management-relevant resolution**.**

We use this framework to address three questions: (1) Can globally learned diversity–environment relationships be transferred to predict population genetic diversity for data-limited regional species? (2) Can meta-learning resolve fine-scale within-species spatial variation more accurately than geographic interpolation alone? (3) How do predicted genetic diversity and projected environmental change combine to shape population and species vulnerability under future climate scenarios? By connecting global genetic information with fine-resolution regional environmental data, our approach provides a scalable framework for translating sparse population genetic observations into multispecies maps of genetic diversity and climate vulnerability.

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2. Method

2.1. Genetic diversity data

Population-level genetic diversity estimates were compiled from two complementary sources. Global records were obtained from the GenDivRange database (Csillery et al., 2025), which contains expected heterozygosity (H_e) and Nei's gene diversity measurements from 19,163 populations representing 1,108 species across diverse taxonomic groups, predominantly estimated from microsatellite markers. When $H_e$was unavailable, Nei's gene diversity was used as a substitute.

Regional data were compiled from the California Population Genomics (CaliPopGen) database (Beninde et al., 2022) and supplemented with expert-contributed records. We restricted the regional dataset to native California coastal species with valid geographic coordinates and population-level genetic diversity estimates. To ensure sufficient spatial and genetic representation, records were retained only when sampling included more than ten individuals per population and, for microsatellite-based studies, at least five genetic markers. Records lacking

valid coordinates or genetic diversity estimates were excluded. Following filtering, the regional dataset comprised 461 populations from 16 marine coastal species (Table S1).

Because available genetic diversity statistics differed among source studies and taxa, species were categorized according to the predominant diversity metric available and modeled as either heterozygosity-type species (H_e) or nucleotide-diversity-type species (π). The California dataset included estimates derived from microsatellites, mitochondrial DNA (mtDNA), and single nucleotide polymorphisms (SNPs) from studies conducted between 1985 and 2020. Metric identity and marker type were retained during data harmonization to account for differences among genetic data sources. The resulting global and regional datasets provided the task distributions for global meta-learning and subsequent transfer to California species within the Conditional Neural Process framework.

2.2. Environmental covariates

We used 23 shared numeric environmental and geographic predictors — geographic coordinates (latitude, longitude, and absolute latitude), nine terrestrial climate variables, three derived climate indices, seven marine environmental variables, and a binary marine indicator — for both global pre-training and California fine-tuning. During California fine-tuning, we included five additional land-use/land-cover predictors (28 predictors total for California). These are described below.

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Terrestrial climate.

Historical terrestrial conditions were represented by nine WorldClim v2.1 bioclimatic variables derived from the 1970–2000 climatological normals (Fick & Hijmans, 2017): annual mean temperature (BIO1), temperature seasonality (BIO4), maximum temperature of the warmest month (BIO5), minimum temperature of the coldest month (BIO6), temperature annual range (BIO7), annual precipitation (BIO12), precipitation of the wettest month (BIO13), precipitation of the driest month (BIO14), and precipitation seasonality (BIO15) (variable definitions following O'Donnell & Ignizio, 2012). Global pre-training populations were characterized at 10-arc-minute resolution (~18.5 km); for the California focal species, we extracted the same variables from 30-arc-second (~1 km) crops to resolve fine-scale coastal gradients. Environmental values were obtained by sampling each raster at the population's coordinates, and missing values at coastal or offshore locations outside terrestrial raster coverage were estimated by nearest-neighbor interpolation from surrounding valid cells.

Derived climate indices

Three indices were computed from the bioclimatic variables to represent characteristics not directly captured by individual variables. Normalized temperature seasonality, BIO4 / (BIO7 × 100 + 1), quantifies temperature variability relative to the annual thermal range (one was added to the denominator to avoid division by zero). Precipitation extremity, BIO13 / (BIO14 + 1), characterizes the strength of seasonal precipitation contrasts, with larger values indicating pronounced wet and dry seasons (one was added to the denominator to avoid undefined ratios where driest-month precipitation approached zero). A composite climate-stress index was computed as $(z(BIO5) - z(BIO14) + z(BIO4)) / 3$, where $z(\cdot)$ denotes standardization to zero

mean and unit variance within each training dataset; higher values indicate environments that are simultaneously hotter, drier, and more thermally variable. Standardizing each component before averaging ensured that temperature and precipitation variables contributed equally despite their differing units and magnitudes.

Marine environment

Marine conditions were obtained from Bio-ORACLE v3.0 (Assis et al., 2024) and comprised the mean, maximum, and annual range of sea-surface temperature, salinity, chlorophyll-a concentration, dissolved oxygen, and ocean pH, representing baseline conditions for ~2000–2019 at approximately 0.05° resolution (~6 km). Values were matched to each population by nearest-neighbor extraction, and populations were classified as marine when the nearest ocean cell lay within 0.5° (~50 km).

Land-use/land-cover pressure (California only). For California, we added five predictors summarizing coastal land-use/land-cover from National Land Cover Database NLCD 2021 (Dewitz, 2023) within a 5-km-radius window around each location: land-use/land-cover pressure (the combined developed and cultivated fraction of land), the developed and cultivated fractions separately, the land fraction of the window, and a binary indicator of NLCD coverage. Land-use/land-cover pressure serves as a proxy for local anthropogenic pressure on the surrounding landscape rather than a direct measurement of habitat disturbance. Because coastal populations lie near or seaward of the shoreline, each window was snapped to the nearest land pixel. Locations south of 32.53 °N (outside the conterminous-US NLCD extent) received no land-cover data and were flagged accordingly, with land-use/land-cover inputs set to zero; global pre-training populations, which lie outside NLCD coverage, likewise carried zeros for these inputs.

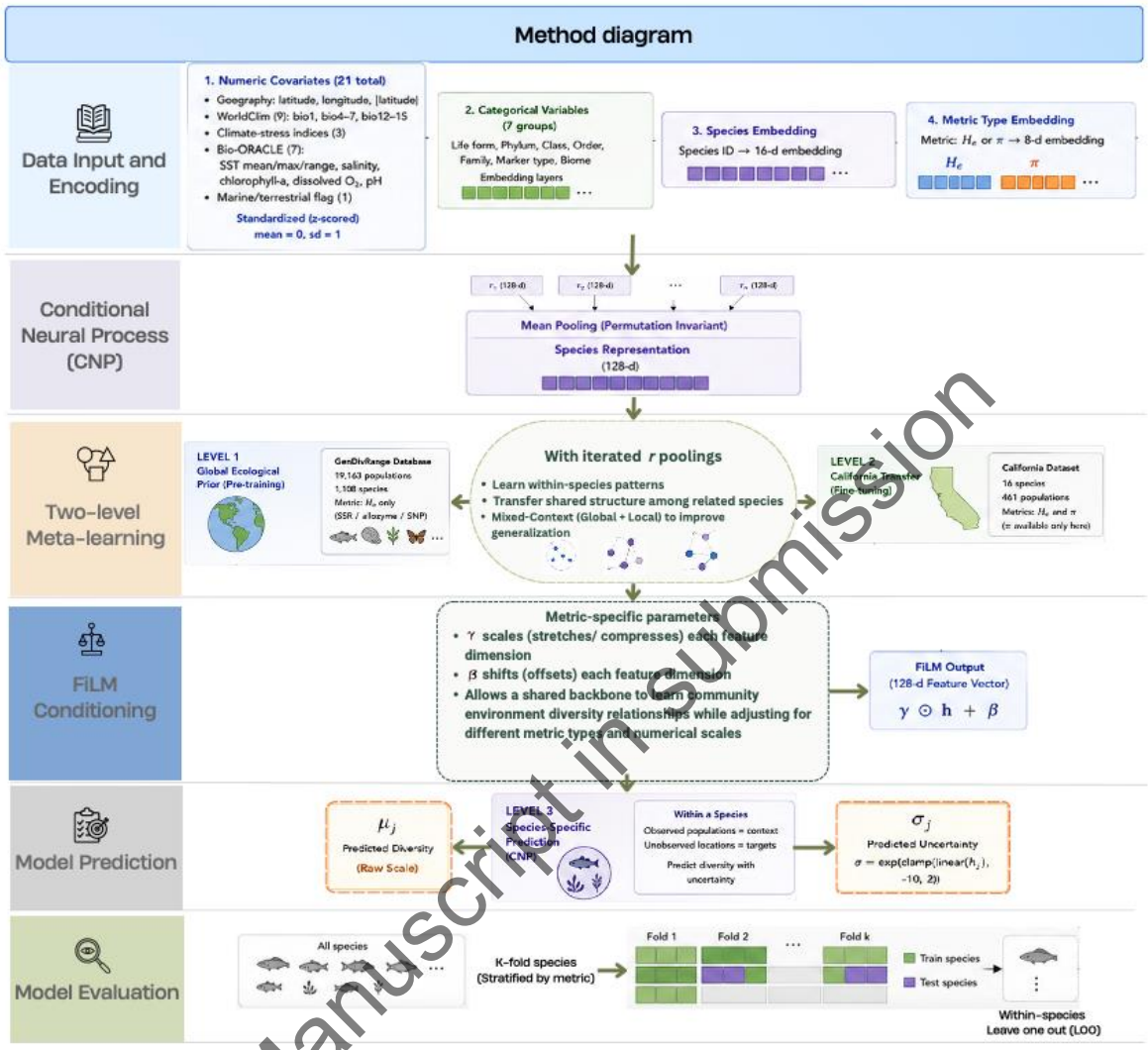
All numeric predictors were standardized to zero mean and unit variance using statistics estimated from the global pre-training data alone and applied unchanged to California, so that the pre-trained feature space was not normalized with knowledge of the downstream domain; California-specific land-use/land-cover predictors were standardized during fine-tuning. We defined categorical and species vocabularies over the combined global and California labels so California taxa and categories carried embedding indices before fine-tuning. Any remaining missing numeric values were set to zero before standardization. We concatenated standardized environmental variables and learned embeddings to form the input representation for the feature encoder.

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Future climate projections.

We projected future environmental conditions under the high-emissions SSP5–8.5 scenario for 2081–2100, intentionally selecting scenarios with the highest environmental variability over time to capture a range of risks. Terrestrial projections used WorldClim v2.1 downscaled CMIP6 output (Fick & Hijmans, 2017; Eyring et al., 2016) at 30-arc-second (~1 km) resolution over California for five global climate models (BCC-CSM2-MR, CanESM5, CNRM-CM6-1, IPSL-CM6A-LR, and MIROC6). We used future marine conditions from Bio-ORACLE v3.0 (Assis et al., 2024) as the mean of the 2080 and 2090 decadal climatologies under SSP5–8.5, matching the WorldClim period; we applied a single marine projection across all five models, so inter-model variation arises from the terrestrial climate. Rather than averaging the models before prediction, we generated projections separately for each global climate model and

summarized them as the ensemble mean, using the between-model standard deviation as a measure of inter-model uncertainty.



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Figure 1. Method diagram of two-phase meta-learning with Feature-wise Linear Modulation (FiLM) conditioning to predict genetic diversity and associated predictive uncertainty.

2.3. Metric-specific target transformations

Expected heterozygosity (H_e) and nucleotide diversity (π) differ in magnitude by roughly two orders of magnitude, so target values were transformed separately by metric before training. H_e was modeled under an identity link and then standardized, yielding a Gaussian predictive distribution in standardized space, consistent with the Gaussian likelihood head of the Conditional Neural Process (Garnelo et al., 2018a). π was first log-transformed as

$\tau = \log(\pi + \varepsilon)$, $\varepsilon = 10^{-4}$,

and then standardized; the offset ε both stabilizes the transform and accommodates the exact-zero π values present in the data. After inverse transformation, π predictions therefore follow a (shifted) lognormal distribution, and the back-transformed median served as the point prediction. Standardization statistics were estimated within each metric — H_e from the global pre-training data and π from the California records, which alone contain π — consistent with the covariate standardization. Predictive intervals were formed by inverse-transforming the standardized-space bounds, so π intervals are asymmetric and strictly positive.

These metric-specific transformations address three limitations of modeling raw diversity values directly. First, standardizing each metric onto a common numerical scale prevents H_e, by virtue of its larger magnitude, from dominating optimization. Second, transformed diversity values enter the context representations on comparable scales, so π observations are not numerically underrepresented during meta-learning. Third, the lognormal formulation enforces the non-negativity of π, eliminating the biologically implausible negative predictions that a Gaussian likelihood on raw values produces.

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2.4. Mean-pooling for context-set aggregation

We modeled genetic diversity with a Conditional Neural Process (CNP; Garnelo et al., 2018a), a member of the neural-process family (Garnelo et al., 2018b). During training, the observed (context) populations of a species are encoded and aggregated into a permutation-invariant latent representation that summarizes the species-specific relationship between environmental conditions and genetic diversity. This representation is combined with the environmental features at each target location to predict genetic diversity and its associated uncertainty. Because the context representation is produced by mean-pooling the encoded observations (a permutation-invariant set operator; Zaheer et al., 2017), predictions are invariant to the order and number of context populations, allowing the framework to accommodate species with widely varying sampling effort.

The CNP aggregates context observations with this mean-pooling operator into a fixed-dimensional, species-level summary. We also evaluated an Attentive Neural Process (ANP; Kim et al., 2019) variant, in which each target location weights context observations by their inferred relevance through an attention mechanism (Vaswani et al., 2017), so that geographically or environmentally similar populations contribute more strongly than dissimilar ones. In our comparisons, however, the mean-pooling CNP outperformed the ANP for California coastal species (Table S/Fig. S): with few sampled populations for most species, the attention mechanism could not learn robust weighting relationships among context observations, whereas the simpler mean-pooling operator acted as an effective regularizer under these data-sparse conditions. We therefore retained the mean-pooling CNP for multispecies prediction of California coastal genetic diversity.

To test whether the covariate-based CNP improved on purely spatial interpolation, we compared it against inverse-distance weighting (IDW; Shepard, 1968), a standard distance-based interpolator that uses neither environmental covariates nor information shared across species. For each species, diversity at a target location was estimated as a weighted mean of that species' observed population values, with weights inversely proportional to great-circle (haversine) distance raised to a power p ($w_i \propto d_i^{-p}$); we used p = 2 and assigned the observed value directly where a target coincided with an observation.

2.5. Two-phase meta-transfer learning framework

To predict population-level genetic diversity across the California coastline, we developed a meta-transfer learning framework based on a Conditional Neural Process (CNP; Garnelo et al., 2018a). Unlike conventional supervised models that fit a separate model per species, the CNP learns a transferable mapping between environmental conditions and genetic diversity across many species by conditioning predictions on a small set of observed populations (the context set) and predicting diversity at unobserved locations (targets). This formulation shares information across species while preserving species-specific responses, making it well suited to datasets in which most species are represented by few populations.

The framework comprises two hierarchical transfer phases (Figure 1), both implemented through episodic, per-species meta-learning tasks in which the observed populations of a species form the context set and held-out populations serve as prediction targets (Vinyals et al., 2016). In the global pre-training phase, the model learns broad relationships between environmental conditions and genetic diversity from the GenDivRange database (Csilléry, 2025; 19,163 populations of 1,108 species, all expected-heterozygosity records), yielding a shared ecological prior across diverse taxa. Pre-training episodes combined within-species tasks — context and targets drawn from the same species — with cross-species tasks that pooled populations of different species within the same taxonomic order, so the model captured both species-level and higher-taxon structure. In the California fine-tuning phase, this prior was adapted to the 16 focal coastal species (populations with both expected heterozygosity and nucleotide diversity), transferring globally learned representations to regional environmental gradients.

Fine-tuning used differential (layer-wise) learning rates (Howard & Ruder, 2018), applying small updates to the pre-trained feature encoder to preserve globally learned environmental representations while more rapidly adapting the species embeddings, the metric-conditioning (FiLM) parameters (Perez et al., 2018), and the decoder to regional conditions. Fine-tuning episodes further included mixed-context tasks that augmented each California species' context set with global populations from the same taxonomic family, allowing ecological information learned during pre-training to inform predictions for species with limited regional sampling.

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2.6. Metric-specific Feature Conditioning

The framework jointly models two measures of genetic diversity, expected heterozygosity (H_e) and nucleotide diversity (π), which differ substantially in numerical scale and statistical properties. Rather than training separate models for each diversity metric, metric identity was incorporated as an additional conditioning variable through Feature-wise Linear Modulation (FiLM). Metric type (H_e or π) was represented by an 8-dimensional embedding, from which two linear transformations generated feature-wise scaling (γ) and shifting (β) parameters. These parameters modulated the latent feature representation produced by the shared multilayer perceptron according to

$$h' = \gamma(m) \odot h + \beta(m),$$

where h denotes the shared latent feature vector and m is the metric embedding. This conditioning allows the shared feature encoder to learn common environment–diversity relationships while accommodating metric-specific differences in variance structure and environmental responses.

2.7. Within-range population genetic diversity prediction

Spatial predictions were generated independently for each species by combining the species-specific context representation with environmental conditions along the California coastal grid. To avoid extrapolation far beyond observed distributions, predictions were spatially restricted using one of two related criteria depending on the analysis. For the coastal transects and gridded per-population predictions, a grid cell was retained if it fell within 2° latitude of at least one observed population of that species (a proximity criterion); cells farther than 2° from any observation were treated as geographic extrapolation. For the resilience assessment, predictions were restricted to each species' occupied latitudinal range extended by a 1.5° buffer at its northern and southern limits, and for species with very narrow observed ranges a minimum extent of approximately five coastal grid cells centered on the species' mean latitude was imposed to ensure sufficient coverage for downstream analyses. Both criteria permit modest interpolation beyond sampled populations while excluding regions the species is unlikely to occupy.

For each species, all observed populations formed the context set: their environmental predictors, species identity, metric type, and observed genetic diversity were encoded and aggregated by permutation-invariant mean pooling into a single latent species representation, which summarizes the environment–diversity relationships learned from that species' observations and was held fixed during prediction. At each target location, the environmental features were combined with this fixed context representation to estimate the predictive mean (μ) and predictive uncertainty (σ) of genetic diversity.

Model performance was evaluated by five-fold cross-validation at the species level (Kohavi, 1995; Roberts et al., 2017), providing a low-variance, approximately unbiased estimate of generalization to unseen species while making full use of the limited California dataset. With only 16 species (461 populations) after filtering, a single train–test split is unreliable, because performance would depend heavily on which two or three species happened to fall in the test set — a particular concern when comparing uncertainty calibration across two diversity metrics with different sample sizes.

Species were partitioned into five approximately equal folds by stratified sampling (Kohavi, 1995) that preserved representation of both expected heterozygosity (H_e) and nucleotide diversity (π) in each fold. In each iteration, one fold was held out and the remaining four were used for California fine-tuning; every fold was fine-tuned from the same global pre-trained model, so that only the regional-adaptation stage differed while the global ecological prior remained fixed. This produced five independently fine-tuned models. Each held-out species was then predicted from its own remaining populations as the context set (i.e., few-shot generalization to a species absent from fine-tuning, rather than zero-shot prediction). Because global pre-training was independent of the California partition, we performed it once, and repeated only the comparatively inexpensive fine-tuning stage per fold. Out-of-fold predictions were pooled across the five folds to yield predictions for all 461 populations, each from a model not trained on that species. Pooled performance was quantified as root mean squared error

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(RMSE), the coefficient of determination (R^2), and empirical coverage of the 68% and 95% predictive intervals, computed separately for H_e and π in raw units.

To distinguish prediction objectives, we report two complementary regimes. Species-level cross-validation measures generalization to entirely new species and provides a conservative bound on cross-species transfer (Roberts et al., 2017). Within-species leave-one-out validation measures interpolation, repeatedly withholding a single population while using the remaining populations of the same species as context; this reflects the framework's primary application — predicting diversity at unsampled locations for partially surveyed species — and was additionally summarized by the Pearson correlation between observed and predicted within-species anomalies.

2.8. Species vulnerability classification

To identify populations potentially susceptible to future environmental change, we computed a composite vulnerability index integrating predicted genetic diversity, a combined environmental stressor, and model uncertainty, following the exposure–sensitivity framing widely used in climate-change vulnerability assessment (Williams et al., 2008; Dawson et al., 2011; Foden et al., 2013):

$$V = (1 - D_norm) \times S + 0.3 \times U_norm,$$

where D_norm is normalized genetic diversity, S is the combined stressor, and U_norm is normalized prediction uncertainty. For the vulnerability index, predicted diversity was normalized **within each species**, rescaling each species' predictions to 0–1 against its own predicted range so that diversity is expressed relative to that species' own variation. The diversity term was inverted (1 − D_norm) so that lower relative diversity contributes more strongly to vulnerability, reflecting the role of standing genetic variation in adaptive potential and evolutionary resilience (Hoffmann & Sgrò, 2011; Sgrò et al., 2011; Pauls et al., 2013), and multiplied by the stressor to reflect that reduced adaptive potential is most consequential under greater environmental stress; the normalized prediction uncertainty was added as a precautionary term (+0.3 U_norm). Predictions were restricted to each species' observed latitudinal range (±1.5° buffer) to limit extrapolation.

The stressor S combined projected climate change with land-use/land-cover pressure, following the cumulative-stressor approach of averaging standardized pressures onto a common scale (Halpern et al., 2008). Climate exposure was defined as the normalized magnitude of projected change under SSP5-8.5 (baseline to 2081–2100), computed as a weighted sum of ten change components normalized against fixed reference-change ranges: marine — change in mean, maximum, and annual range of sea-surface temperature, ocean acidification (ΔpH), deoxygenation (ΔO_2), and thermal novelty (warming relative to the baseline annual SST range) — and terrestrial — change in maximum temperature of the warmest month (ΔBIO5), annual mean temperature (ΔBIO1), temperature seasonality (ΔBIO4), and driest-month precipitation. Marine species used all components; terrestrial-only species used the terrestrial components renormalized to sum to one (component weights in Table S). For populations within NLCD coverage, land-use/land-cover pressure — the developed-plus-cultivated fraction of land within a 5-km window, a proxy for local anthropogenic pressure (Venter et al., 2016) — was combined

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with climate exposure as $S = (1 - w)\cdot E + w\cdot LULC$, with $w = 0.25$; elsewhere S equaled climate exposure alone.

Prediction uncertainty was the CNP predictive standard deviation, normalized within species to 0–1. The uncertainty term ($0.3 \times U_norm$) up-weights populations whose predictions are less certain, typically in data-poor regions. The coefficient 0.3 was not empirically calibrated; it is a moderate penalty, sufficient to shift borderline populations between resilience classes but not to override strong signals from the diversity–stressor interaction.

2.9. Aggregated population-level vulnerability

To summarize multispecies genetic vulnerability to acquire ecosystem-level vulnerability along the California coast while giving equal weight to species with different spatial distributions, we applied a deterministic sliding-window aggregation to species-specific vulnerability predictions. Vulnerability was available for each of 16 species at coastal pixels within its modeled range across a 500-pixel coastal grid spaced approximately 3 km apart. Pixel locations were converted to cumulative along-coast distance using great-circle distances between consecutive pixels, allowing spatial aggregation to be defined by shoreline distance rather than geographic degrees.

We evaluated vulnerability at two spatial scales using 10-km and 25-km sliding windows, centered on each coastal pixel and including pixels within ±5 km and ±12.5 km, respectively. Within each window, vulnerability was first averaged across pixels separately for each species, producing one mean vulnerability estimate per species present. These species-specific means were then averaged with equal weight to obtain the window-level vulnerability, preventing species represented by more pixels within a window from disproportionately influencing the estimate.

Final coastal vulnerability at each pixel was calculated as the mean of all window-level estimates whose centers occurred within half the specified window width of that pixel (±5 km or 12.5 km). This second averaging reduced abrupt variation arising from individual four-species draws while retaining local spatial structure. The resulting surface therefore represents a locally species-weighted estimate of multispecies vulnerability with standardized sampling effort. The sliding-window surface is a one-dimensional mainland coastal transect and therefore does not cover the offshore Channel Islands. To place the eight islands on the same vulnerability scale, each island was assigned a single mean vulnerability computed at its centroid

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3. Results

3.1. Meta-learning performance

The Conditional Neural Process showed strong predictive performance across both global pre-training and California fine-tuning datasets (Figure 2). For the 16 California focal species, within-species leave-one-out predictions closely matched observed genetic diversity for both diversity metrics. Expected heterozygosity (He) showed particularly strong agreement between observed and predicted values (R^2=0.96), with the predicted spread corresponding to 96% of the observed spread (Figure 2a). Predictions closely followed the 1:1 relationship across the observed He range, indicating little systematic compression toward species means. Nucleotide diversity (π) was also predicted with relatively high accuracy (R^2=0.74), although performance was lower than for He (Figure 2b). Predictions reproduced the dense concentration of low-π

observations but showed greater uncertainty and some compression at the upper end of the distribution, with the predicted spread corresponding to approximately 75% of the observed spread. This pattern was most evident for high-π populations of *Emerita analoga*, where predicted means were lower than observed values and accompanied by relatively broad predictive intervals. Thus, although the metric-specific transformation and likelihood substantially improved prediction of nucleotide diversity, extreme π values remained more difficult to resolve than He.

Because pooled prediction statistics can be dominated by differences in mean genetic diversity among species, we separately evaluated whether the model captured population-level variation within species. After centering observed and predicted values relative to their respective species means and expressing these deviations in standard-deviation units, observed and predicted within-species anomalies remained strongly correlated (pooled r=0.83; Fig. Xc). The median species-level correlation was r=0.80, and all 16 focal species exhibited positive observed–predicted correlations. These results demonstrate that predictive skill was not attributable solely to distinguishing species with high versus low mean genetic diversity; the model also recovered substantial spatial variation among populations within species.

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Figure 2. Observed versus predicted genetic diversity. (a, b) Leave-one-out predictions for the 16 California focal species (μ ± 95% CI; 1:1 dashed): expected heterozygosity (He, $R^2 = 0.96$) and

nucleotide diversity (π, $R^2 = 0.74$). (c) Within-species anomalies (deviation from each species' mean, SD units), isolating spatial from between-species variation — the model resolves within-species structure across all taxa (median per-species r = 0.80; 0/16 negative). (d) Global held-out predictions from pre-training (n = 6,245 populations, 1,108 species; $R^2 = 0.81$), showing the diversity–environment relationship generalizes worldwide.

Global pre-training provided a transferable foundation for regional fine-tuning. Predictions for 6,245 held-out global populations spanning 1,108 species closely corresponded with observed He (R2=0.81, RMSE = 0.098; Figure 2), indicating that the environment–diversity relationships learned during pre-training generalized across geographically and taxonomically diverse populations. These globally-learned representations subsequently provided the initialization for California fine-tuning, where the model adapted to regional environmental conditions and species-specific diversity patterns.

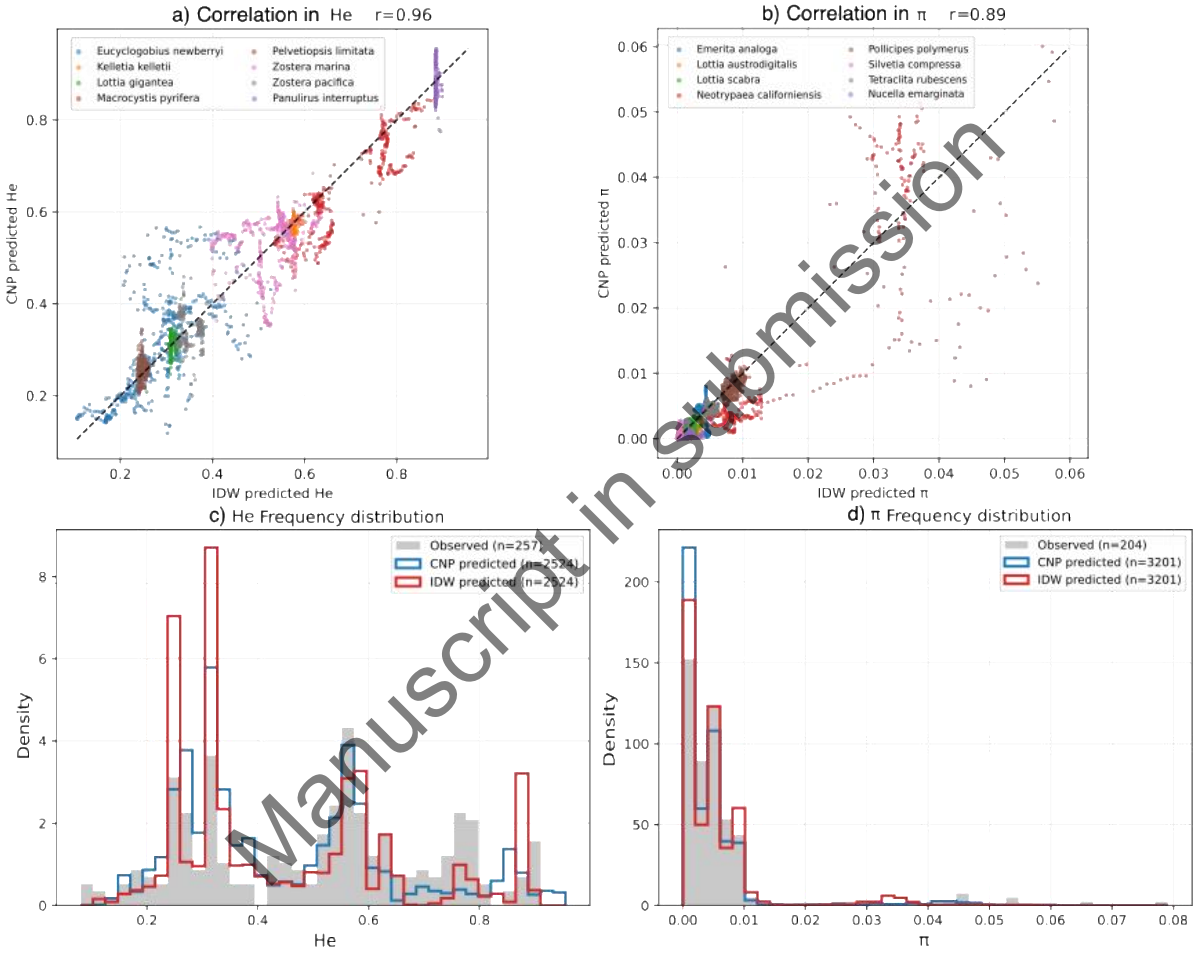
These evaluations demonstrate predictive skill at complementary scales: global pre-training captured broad, transferable relationships between environmental conditions and genetic diversity, whereas California fine-tuning resolved both among-species differences and within-species population-level variation. Performance was stronger for He than for π, particularly at the upper tail of the nucleotide-diversity distribution. Because the California evaluation used leave-one-population-out predictions for species retained during fine-tuning, these statistics characterize interpolation within the focal species rather than generalization to previously unseen taxa. Calibration analyses indicated that empirical coverage closely matched nominal coverage for prediction intervals up to approximately 50–60%, after which undercoverage became increasingly apparent. Nominal 95% prediction intervals captured approximately 89% of held-out observations, indicating modest underestimation of uncertainty in the tails of the predictive distribution. Calibration was strongest within the central prediction range, whereas interval-based assessments at higher confidence levels should be interpreted with greater caution. However, species-level analyses indicated that predictive performance was driven primarily by differences among species rather than by within-species spatial variation. Most individual species exhibited low or negative leave-one-out R^2 values, suggesting that the model more effectively captured broad taxonomic differences in genetic diversity than fine-scale population-level gradients within species.

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3.2. CNP model comparison with inverse-distance weighting

Predicted genetic diversity from the Conditional Neural Process (CNP) and from inverse-distance weighting (IDW) was strongly correlated across the coastal grid (Figure 3a; expected heterozygosity r = 0.96, nucleotide diversity r = 0.89), with a negligible mean difference between methods (He, −0.001; π, −0.0007) and each species occupying a distinct band along the 1:1 line; the largest between-method departures were confined to sparsely or unevenly sampled species (e.g., *Eucyclogobius newberryi*, *Neotrypaea californiensis*). The two methods also produced similar overall diversity distributions (Figure 3b), differing mainly in form: being distance-based, IDW concentrated predicted values near the sampled—often spatially clustered—observations, whereas the covariate-based CNP produced a spatially continuous surface. This difference in form is not itself evidence that either surface is more accurate. The close agreement in both correlation and distribution indicates that the coarse spatial structure of diversity, and the resulting vulnerability rankings, are robust to the choice of interpolation method. The CNP's

advantages over IDW are strongest where they can be directly tested. Within the studied species, it resolved within-species spatial structure substantially better (within-species anomaly r: He 0.79 vs 0.91, π 0.43 vs 0.74), predicted nucleotide diversity more accurately (R^2 0.47 vs 0.78), and provided calibrated predictive uncertainty, which IDW does not. As a covariate-based model it can additionally, in principle, extrapolate along environmental gradients beyond a species' sampled range and share information across species—capabilities a distance-based interpolator lacks—but our species-level cross-validation shows these do not translate into reliable prediction for entirely unsampled species (leave-species-out π R^2 = 0.06). These should therefore be described as structural properties of the model rather than demonstrated gains.



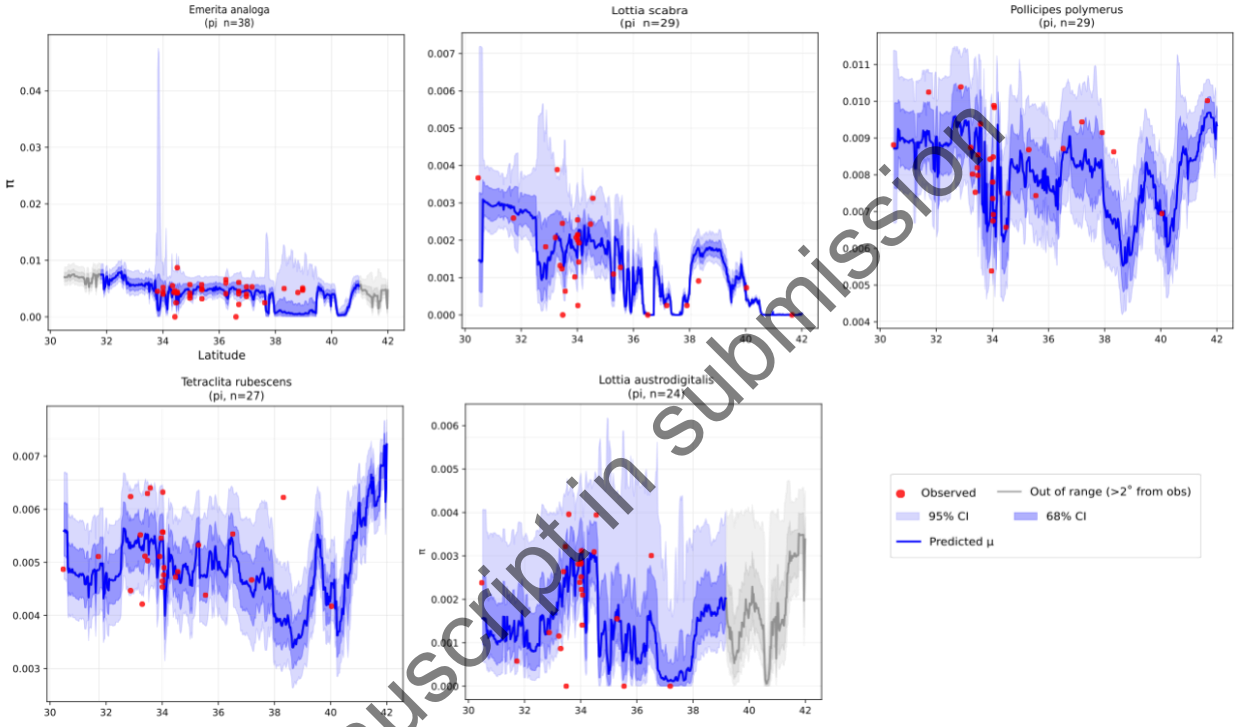
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Figure 3. Comparison between Conditional Neural Process (CNP) and inverse-distance weighting (IDW) based on (a-b) Correlation in predicted expected heterozygosity (He) and nucleotide diversity (π) by both methods, and (c-d) Frequency Distribution of predicted He and π

3.3. **California coast genetic diversity and climate exposure spatial distribution**

Figure 4 shows genetic diversity prediction for Emerita analoga ($n = 38$), Lottia scabra ($n = 29$), Pollicipes polymerus ($n = 29$), Tetraclita rubescens ($n = 27$), and Lottia austrodigitalis ($n = 24$). Predictions are displayed against latitude for visualization, but spatial

variation is generated from the environmental covariates associated with each coastal grid location rather than latitude alone. Predicted genetic diversity exhibited substantial species-specific spatial structure along the California coast (30.5–42°N; Figure 4 and 5). In contrast to broad differences in mean diversity among species, the coastal transects revealed fine-scale variation within species that closely corresponded with observed population-level patterns. This spatial structure is consistent with the strong within-species predictive correspondence identified in the leave-one-out evaluation (median species-level r=0.80; Figure 4). Predictive intervals generally widened in areas with fewer nearby observations and toward species' sampled range limits



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Figure 4. Latitudinal transects of predicted nucleotide diversity (π) for California coastal species. For each species, the Conditional Neural Process (CNP) predicted mean π(dark blue line) is shown across latitude along the California coastal grid, with 68% (darker shading) and 95% (lighter shading) predictive intervals representing CNP predictive uncertainty. Red points indicate observed population-level π, with sample size (n) reported for each species. Grey lines and shading identify portions of the transect more than 2° latitude from the nearest observed population, where predictions are less constrained by local genetic observations and should be interpreted cautiously

Several taxa exhibited reductions in predicted diversity along portions of the central and northern California coast, particularly around 37–38°N. The recurrence of these patterns among taxonomically and ecologically distinct species suggests that shared spatial environmental gradients may contribute to geographic structuring of genetic diversity. However, the magnitude and location of these transitions differed among species, indicating that common environmental gradients interact with species-specific ecological and demographic processes rather than producing a uniform coast-wide response.

Nucleotide diversity (π) occurred at a much smaller absolute scale but nevertheless exhibited detectable spatial structure (Figure 4). Pollicipes polymerus maintained comparatively high π across much of its sampled range, with localized reductions along the central coast, whereas Emerita analoga and Tetraclita rubescens exhibited reductions toward approximately 38–40°N. Both Lottia scabra and L. austrodigitalis showed broader declines from southern toward northern portions of their sampled distributions. Uncertainty was generally greater relative to the magnitude of π than for He, and the highest observed values of E. analoga remained underpredicted, consistent with the upper-tail compression identified in the model evaluation.

Spatial structure in predicted genetic diversity was particularly evident among species characterized by expected heterozygosity (He; Figure 5, a-d), although its magnitude and direction differed among species. Each panel shows the CNP posterior mean He (blue line) against latitude with 68% (darker) and 95% (lighter) prediction intervals and observed population estimates (red points); transect segments more than 2° latitude beyond a species' observed range are drawn in grey to denote extrapolation and its elevated uncertainty.

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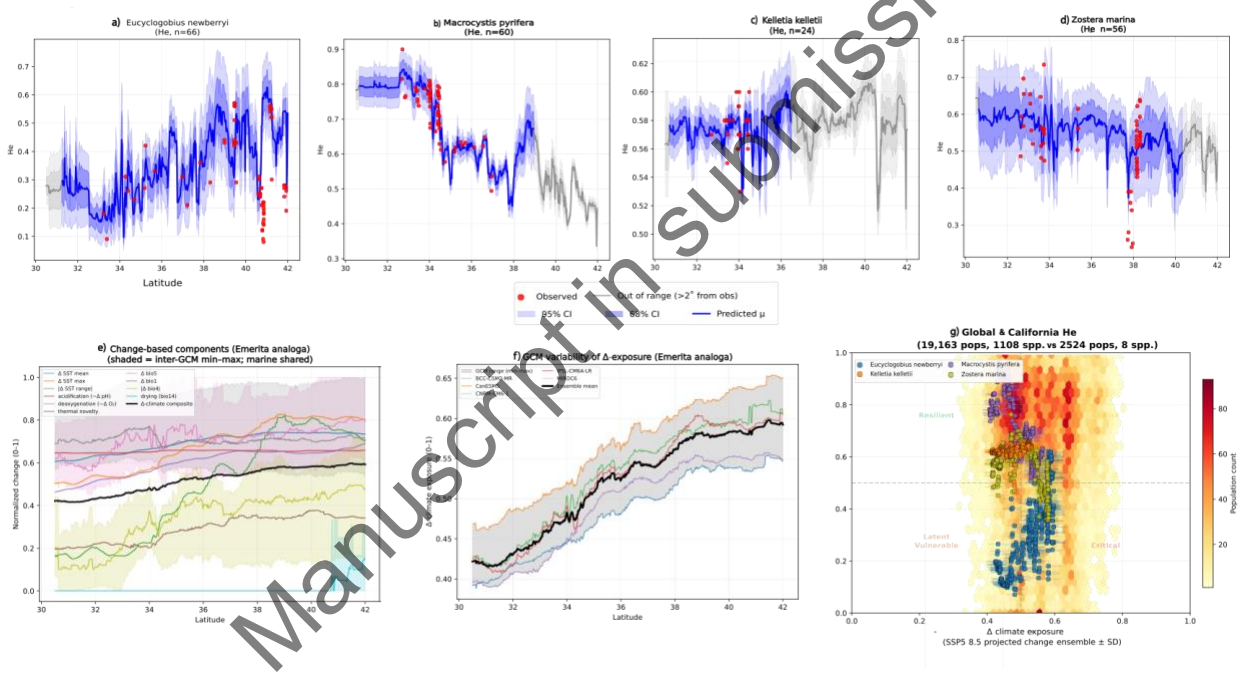


Figure 5. Predicted expected heterozygosity (He), climate-change exposure, and their joint distribution for California coastal species. (a-d) Latitudinal transects of predicted He for four He species — Eucyclogobius newberryi, Macrocystis pyrifera, Kelletia kelletii, and Zostera marina. (e) Decomposition of the Δ-climate exposure index into its normalized (0–1) change-based components for an exemplar species (Emerita analoga); (f) Between-GCM variability of the Δ-climate exposure composite for the same exemplar: each of the five GCMs (colored), the ensemble mean (black), and the GCM min–max range (grey). (g) Joint diversity–exposure space: the global population cloud (19,163 populations, 1,108 species) as a hexbin density (color = population count) in He (y-axis, 0–1) versus Δ-climate exposure (x-axis; SSP5–8.5 projected change, 2081–2100), with four California He species overlaid as points colored by species and horizontal bars giving the ensemble mean ± between-GCM SD of exposure. Dashed lines at 0.5

partition the space into diversity–exposure regions — Resilient (high diversity, low exposure), At Risk (high diversity, high exposure), Latent Vulnerability (low diversity, low exposure), and Critical (low diversity, high exposure).

Macrocystis pyrifera retained relatively high diversity across its sampled range but declined from southern to central California — a high-diversity plateau south of ~33°N giving way to a sharp reduction near 34°N — with predictions closely tracking observed population-level variation. Zostera marina showed stronger localized structure, including a marked reduction in predicted He near 37–38°N followed by partial recovery farther north. Eucyclogobius newberryi exhibited pronounced fine-scale variation superimposed on a broader latitudinal gradient, including localized reductions toward the northern portion of its sampled range. In contrast, Kelletia kelletii was sampled almost entirely in southern California, so predictions farther north rested on extrapolation and carried substantially greater uncertainty. Across species, areas of high projected climate exposure did not consistently coincide with areas of low genetic diversity.

Projected climate exposure showed a clear latitudinal gradient along the California coast, increasing from approximately 0.42 in southern California to 0.60 in northern California (ensemble mean; Figure 5e). This spatial pattern was broadly consistent across the five GCMs, with relatively limited inter-model variation (approximately 0.02–0.04) (Figure 5f). Variation among GCMs arose primarily from the terrestrial climate components, whereas marine components were derived from a common Bio-ORACLE projection.

The joint diversity–exposure assessment (Figure 5g) therefore revealed contrasting vulnerability among populations experiencing comparable levels of projected climate change. Terrestrial drivers (Δbio1, |Δbio4|, Δbio5, drying = Δbio14) and marine drivers (ΔSST mean, ΔSST max, |ΔSST range|, acidification = −ΔpH, deoxygenation = $-\Delta O_2$, thermal novelty), with the aggregate Δ-climate composite (black); shading is the inter-GCM min–max range (marine layers shared across GCMs). Populations that combined relatively low genetic diversity with high projected exposure occupied the Critical region of diversity–exposure space, whereas populations exposed to similarly high change but retaining greater diversity fell within the “At Risk” region; those with low diversity but lower projected exposure occupied the Latent Vulnerability region.

Fine-scale fluctuations in predicted diversity reflected spatial variation in the environmental and anthropogenic predictors used by the model, including high-resolution climate and land-use conditions. Although the leave-one-out analysis demonstrates that the model captures population-level variation within species, individual fine-scale peaks and troughs should not necessarily be interpreted as discrete biological boundaries because their spatial resolution may exceed that of the available genetic observations. Accordingly, the transects are most informative for identifying broader spatial gradients and recurrent regions of change, while localized features require additional population-level sampling for validation.
Projected climate exposure and genetic diversity jointly structure population vulnerability

3.4. Projected changes in genetic diversity under future climate

Under SSP5–8.5, projected changes in genetic diversity by 2081–2100 were generally modest and varied in both magnitude and direction among species (Figure 6). Among species

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characterized by expected heterozygosity (He), Zostera marina exhibited the largest decline, from approximately 0.52 under historical conditions to 0.43 under future conditions. Smaller declines were projected for Pelvetiopsis limitata, Panulirus interruptus, Macrocystis pyrifera, Kelletia kelletii, and Lottia gigantea. In contrast, Eucyclogobius newberryi increased from approximately 0.32 to 0.39, and Zostera pacifica from approximately 0.35 to 0.39. Projected changes in nucleotide diversity (π) were smaller in absolute magnitude and similarly heterogeneous, with increases for some species, including Neotrypaea californiensis and Lottia scabra, and decreases for others, including Emerita analoga and Tetraclita rubescens. For several species, variation among GCM projections overlapped the historical estimate, indicating uncertainty in the direction of projected change.

At the multispecies level, these contrasting responses largely offset one another. Mean normalized genetic diversity changed only slightly, from 0.265 under historical conditions to 0.263 under future conditions (between-GCM SD = 0.005). The latitudinal pattern of community-level diversity was also largely retained, with the future ensemble closely following the historical profile across much of the California coast. Thus, although individual taxa exhibited potentially meaningful changes, particularly the decline projected for Z. marina, the model did not indicate a consistent coast-wide directional decline in genetic diversity.

Between-GCM uncertainty was modest across most of the California coast but increased toward northern latitudes, with the largest variation occurring at approximately 40–42°N. This pattern indicates that projected genetic diversity in northern populations was more sensitive to differences among the five terrestrial climate projections, consistent with the greater magnitude and inter-model variability of projected environmental change at higher latitudes. Nevertheless, between-GCM variation remained small relative to the overall spatial variation in genetic diversity, and the historical and future community-level profiles remained broadly similar across models. Importantly, this variation captures only uncertainty associated with GCM choice; it is distinct from the CNP's predictive uncertainty and does not encompass uncertainty arising from model structure, genetic sampling, or the shared future marine projections.

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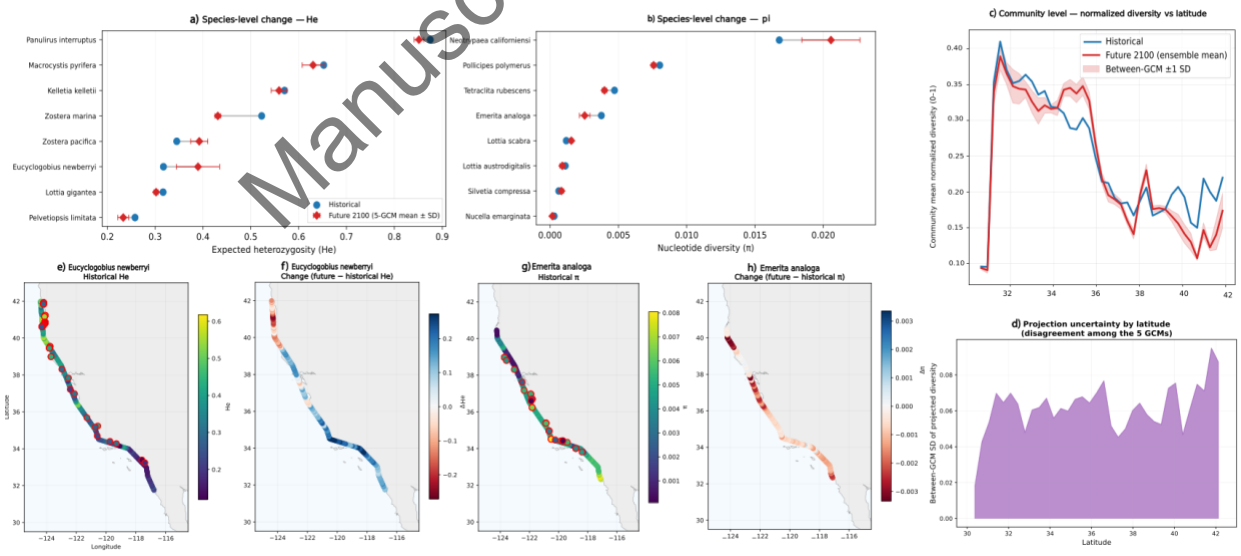


Figure 6: Projected changes in population genetic diversity of California coastal species under SSP5–8.5 (2081–2100). (a, b) Historical and future range-mean genetic diversity for species

measured using (a) expected heterozygosity (He) and (b) nucleotide diversity (π). Blue circles indicate historical predictions and red diamonds indicate future ensemble means; error bars indicate ±1 SD among GCMs. (c) Historical and future community-level normalized genetic diversity along latitude. The coast-wide normalized mean changed from 0.265 historically to 0.263 ± 0.005 under future conditions. (d) Latitudinal variation in between-GCM SD of projected diversity. (e–h) Spatial examples showing historical diversity and projected change (future − historical) for Eucyclogobius newberryi (He; e–f) and Emerita analoga (π; g–h); outlined points indicate observed populations. Overall, projected changes were modest and mixed in direction among species, producing little net change in community-mean diversity despite spatially structured responses within individual taxa.

Spatially explicit projections further showed that range-averaged species responses reflected coherent geographic changes rather than simply averaging opposing local responses. For Eucyclogobius newberryi, historical He varied substantially across its modeled coastal range, but projected changes were predominantly positive, consistent with the increase in range-mean He from approximately 0.32 to 0.39. In contrast, Emerita analoga exhibited relatively high historical π in southern portions of its range but predominantly negative projected changes across its modeled distribution, consistent with its decline in range-mean π. These examples demonstrate that species exposed to the same regional climate-change scenario can exhibit contrasting genetic-diversity responses, depending on their learned diversity–environment relationships and spatial environmental context. Consequently, the near stability of community-mean diversity (0.265 historically versus 0.263 in the future) should not be interpreted as stability at the individual-species or population level. Rather, it emerges from offsetting, spatially structured increases and decreases among taxa, which can conceal potentially important species-level changes when genetic diversity is summarized at the community scale.

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3.5. Climate vulnerability along the California Coast

Present-day multispecies genetic vulnerability to projected climate change under SSP5–8.5 (2081–2100) showed a patchy spatial distribution across 16 California coastal species and 5,725 population-level predictions (Figure 7). These high vulnerability areas represent regions where many of the sampled species have low intraspecific genetic diversity with high climate exposure. The 10-km window emphasized finer-scale local variation and sharper vulnerability hotspots, whereas the 25-km window produced a smoother representation of broader regional patterns.

Vulnerability, calculated from within-species diversity deficit, projected climate exposure, and predictive uncertainty, increased from approximately 0.30 in southern California (32.54–34.43°N), compared with approximately 0.40–0.48 in northern California (≥37.82°N). The diversity deficit (1 − normalized diversity) was the largest and most spatially variable component, increasing overall toward higher latitudes while producing much of the local variation in vulnerability. In contrast, projected climate exposure increased more smoothly from south to north, from approximately 0.42 to 0.59, with a narrow inter-model spread across the five-GCM ensemble (BCC-CSM2-MR, CanESM5, CNRM-CM6-1, IPSL-CM6A-LR, and MIROC6). The resulting vulnerability estimates also showed a narrow range among GCMs, indicating that the broad latitudinal pattern was relatively insensitive to climate-model choice.

Spatially, ensemble-mean vulnerability identified high-vulnerability areas in the San Francisco–Marin region, Monterey, the Santa Barbara Channel, and the far-northern coast near Del Norte. Across the eight Channel Islands, predicted genetic climate vulnerability was uniformly moderate and tightly clustered, ranging from 0.253 (San Clemente) to 0.294 (Santa Cruz) with a mean of 0.274 (Supplementary Information Table S2). These patterns were consistent between the 10-km and 25-km sliding-window analyses, which used species-equal averaging within each window. The 10-km window resolved sharper and more localized maxima, whereas the 25-km window produced smoother, broader regional patterns without substantially changing hotspot locations. Thus, the principal geographic patterns were robust to spatial aggregation scale. Overall, present-day genetic vulnerability was structured primarily by geographic variation in relative genetic diversity and amplified by projected climate exposure, with the major latitudinal gradient and spatial hotspots remaining consistent across both GCM projections and spatial smoothing scales.

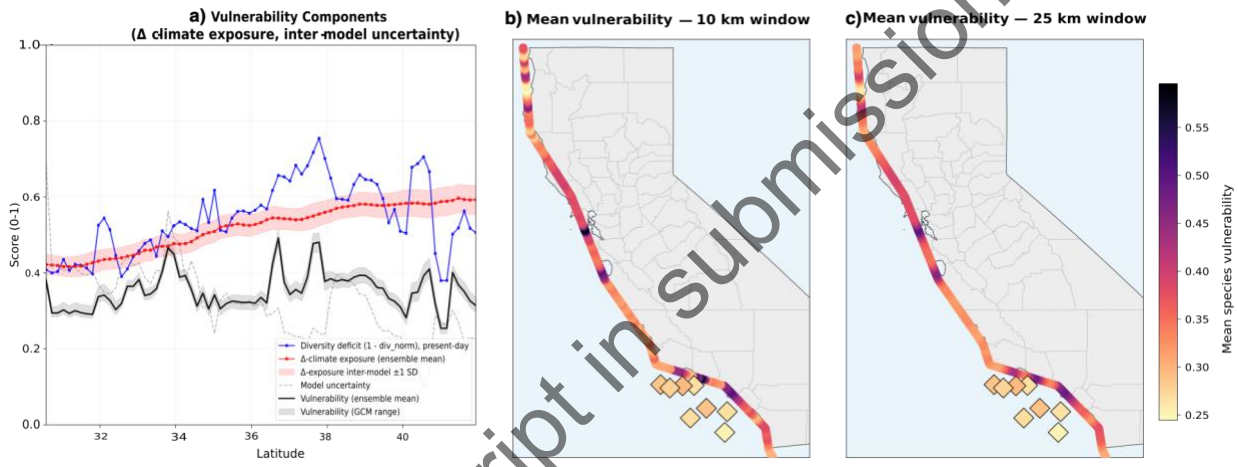


Figure 7. Ecosystem-level genetic vulnerability of California coastal species to projected climate change (SSP5–8.5, 2081–2100; 5-GCM ensemble; 16 species, 5,725 species-level predictions). (a) Latitudinal profiles of vulnerability and its components; shaded bands indicate variation among GCMs. (b, c) Ensemble-mean vulnerability mapped with a deterministic sliding-window mean at (b) 10 km and (c) 25 km windows (shared color scale)

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4. Discussion

From a conservation perspective, our results show the potential to leverage globally available genetic datasets to inform local and regional conservation planning where direct genetic sampling remains limited (Miraldo et al., 2016; Manel et al., 2020). Because the pretraining dataset included only four California species, successful prediction across the remaining species indicates that broad ecological and evolutionary relationships governing genetic diversity can transfer across taxa and regions (Theodoridis et al., 2020; Manel et al., 2020). This capability is particularly relevant for landscape-scale conservation, where comprehensive genetic surveys are often prohibitively expensive or logistically infeasible (Shafer et al., 2015). By combining sparse local observations with information learned from global biodiversity datasets, the framework provides a mechanism for generating continuous maps of genetic resilience across entire coastlines or management regions. Such products can

help identify populations that may be vulnerable to future environmental change, prioritize areas for additional monitoring, and incorporate evolutionary resilience into ecosystem-based management and climate adaptation planning (Sgrò et al., 2011; Hoban et al., 2020; Funk et al., 2012).

From a methodological perspective, the modest undercoverage observed at higher confidence levels indicates that predictive uncertainty was slightly underestimated for extreme observations, a behavior commonly reported for Gaussian-output neural process models (Garnelo et al., 2018; Guo et al., 2017). Future work could improve uncertainty quantification through post-hoc recalibration, calibration-aware loss functions, or the use of heavier-tailed predictive distributions such as the Student-t (Kuleshov et al., 2018; Guo et al., 2017).

Several approaches may also improve prediction of nucleotide diversity. Transforming π values prior to modeling (e.g., $\log(\pi + c)$, where c is a small constant) may increase effective dynamic range, although sensitivity analyses would be required given the prevalence of near-zero values. Metric-specific models for H_e and π, as well as separate treatment of species exhibiting near-constant diversity across populations, may further improve predictive performance while retaining the advantages of the meta-learning framework.

Ecosystem vulnerability masks contrasting species-specific responses: declines in some taxa were counterbalanced by increases or relative stability in others (Bálint et al., 2011; Pauls et al., 2013). This distinction between species-level and ecosystem-level vulnerability is important because a nearly unchanged multispecies mean can coexist with substantial redistribution of genetic diversity among taxa and locations. *Z. marina*, for example, showed a decline of approximately 0.09 in π, whereas *E. newberryi* and *Z. pacifica* showed increases, illustrating that projected responses depend strongly on species-specific relationships with environmental change (Reusch et al., 2005).

The projections should nevertheless be interpreted as scenario-dependent estimates rather than deterministic forecasts. Future environmental conditions extend beyond the temporal domain represented during model training, and for several species the magnitude of projected change was comparable to variation among GCMs (Williams et al., 2007; Fitzpatrick & Hargrove, 2009). Moreover, the displayed future uncertainty represents between-GCM variability rather than total predictive uncertainty (Hawkins & Sutton, 2009): it does not incorporate CNP predictive uncertainty or uncertainty associated with model structure. Because the marine covariates were derived from a common Bio-ORACLE SSP5–8.5 projection (Assis et al., 2018), the between-GCM variation shown here primarily reflects differences among terrestrial climate projections (Fick & Hijmans, 2017). Consequently, confidence is greater in the overall result of heterogeneous responses than in the precise magnitude or direction projected for any individual species.

The spatial pattern of projected resilience, a combined metric of climate change and population genetic diversity, reveals information that would be obscured if we considered either dimension independently (Williams et al., 2008; Dawson et al., 2011; Foden et al., 2013). Northern California generally experiences greater projected environmental change, but this does not translate uniformly into greater vulnerability because populations differ substantially in their underlying diversity.

Coastal genetic vulnerability showed a broad northward increase, but this pattern was driven more strongly by geographic variation in genetic diversity than by projected climate exposure alone. The diversity deficit was the largest and most spatially variable component of vulnerability, generating much of the local structure along the coast, whereas projected exposure

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increased more gradually from south to north. This distinction suggests that populations exposed to similar levels of environmental change may differ substantially in vulnerability depending on the genetic diversity they currently retain. Populations with relatively high diversity may have greater potential to respond to changing environmental conditions (Barrett & Schluter, 2008; Hoffmann & Sgrò, 2011), whereas the coincidence of low diversity and high exposure produced pronounced vulnerability hotspots, particularly around San Francisco–Marin, Monterey, the Santa Barbara Channel and Channel Islands, and the far-northern coast. The persistence of these vulnerability hotspots—particularly around San Francisco–Marin, Monterey, the Santa Barbara Channel/Channel Islands, and the far-northern coast—across both 10- and 25-km spatial windows, together with the narrow variation among GCMs, indicates that the broad spatial pattern is primarily associated with underlying variation in genetic diversity and exposure rather than climate-model or spatial aggregation choice. The results therefore show that geographic exposure alone does not determine vulnerability; instead, projected environmental change interacts with existing spatial variation in genetic diversity to produce geographically heterogeneous capacity to respond to future conditions (Foden et al., 2013; Bay et al., 2018; Razgour et al., 2019).

5. Conclusion

We developed a Conditional Neural Process meta-learning framework that transfers globally learned genetic diversity–environment relationships to data-limited regional species, enabling sparse population genetic observations to be downscaled into fine-resolution, spatially continuous multispecies predictions. Applied to 16 California coastal species, the framework reproduced observed diversity well, with stronger performance for expected heterozygosity He (R^2=0.96) than nucleotide diversity (π) (R^2=0.74). More importantly, the model captured within-species spatial variation across all focal species (median species-level (r=0.80), demonstrating the predictive skill required to resolve population-level patterns rather than simply differences among species. Global pre-training provided a transferable foundation for prediction with limited regional data, while predictive uncertainty appropriately increased for less well-constrained observations, particularly at the upper range of π. Together, these results support global-to-local meta-learning as an approach for bridging the mismatch between sparse genetic sampling and the fine spatial scales relevant to conservation.(maybe something here about how it performed better than IDW interpolation)

Across the California coast, genetic vulnerability showed a broad northward increase, but its spatial structure was driven more strongly by variation in relative genetic diversity than by projected environmental exposure alone. The diversity deficit generated much of the local variation in vulnerability, whereas projected exposure increased more gradually from south to north. This interaction produced persistent vulnerability hotspots around San Francisco–Marin, Monterey, the Santa Barbara Channel/Channel Islands, and the far-northern coast**.** These broad spatial patterns were consistent across five GCMs and remained evident at both 10- and 25-km aggregation scales, indicating that the principal geographic patterns were relatively robust to climate-model choice and spatial smoothing.

More broadly, our framework provides a scalable approach for transforming sparse genetic observations into fine-resolution assessments of genetic diversity and climate vulnerability. These predictions remain conditional on sampling coverage, contemporary diversity environment relationships, and future environmental scenarios and should complement

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rather than replace empirical genetic monitoring. By identifying where low relative diversity coincides with high projected exposure, where individual populations depart from broader species-level patterns, and where predictions remain uncertain, global-to-local downscaling can help prioritize targeted genetic sampling and integrate population-level genetic variation into spatial conservation and climate-adaptation planning.

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